

TranscendiantRouter

Every token earned. Most answers free.

AMD Developer Hackathon ACT II | Track 1: General-Purpose AI Agent

Team Transcendiant | solo build by David Castro

THE GAME

Scoring rewards the frugal, not the flashy

- Every answer is judged for accuracy. Below the gate, you are out.
- Everyone who passes is ranked by ONE thing: fewest Fireworks tokens.
- Local model tokens count as ZERO.
- So the winner is whoever knows exactly when NOT to pay.

THE INSIGHT

Route by measured weakness, not by vibes

- A strong small model is FREE. Qwen3-4B answers most categories perfectly.
- Measure where it is weak, and pay the expert for exactly that.
- Confidence is checked per answer: an explicit final answer for math,
 - a compile check for code, self-consistency voting everywhere else.
- An answer that cannot prove itself escalates. Nothing else does.

ARCHITECTURE

Classify, answer locally, escalate on doubt

- 1. Rule-based classifier sorts each task into 8 categories. Zero tokens.
- 2. Qwen3-4B (llama.cpp, weights baked in) answers locally on 2 CPU cores.
- 3. Answers that prove themselves ship free: math with a final-answer check,
 - code that parses, consistent votes elsewhere.
- 4. Logical reasoning and unproven answers escalate to kimi-k2p7-code.

HOSTILE ENVIRONMENT

The judge's proxy is discovered, never assumed

- A boot-time probe sweeps base URLs, model ID forms, transports, and
- auth shapes, then pins the first combination that answers.
- Model IDs resolve from ALLOWED_MODELS at runtime. Nothing hardcoded.
- Off-list calls are impossible by construction.
- If nothing answers, the router degrades to local-only and still finishes.

TIME AS AN ADVERSARY

A timeout scores zero, so timeouts are impossible

- Time-fit preemption: a local generation that cannot finish inside the remaining budget never starts. It escalates instead.
- Results are checkpointed atomically after every task: even a hard kill leaves a valid, scoreable results.json.
- Cost adapts to the host: fast box ~331 tokens, degraded box ~2,600.

MEASUREMENT

A 272-task harness, built before the router

- GSM8K math, HumanEval code, and authored tasks across all 8 categories.
- Held-out split never touched by tuning. Generated code actually executes.
- Offline replay: record models once, re-score every routing policy instantly.
- Every threshold in the shipped image comes from a measured rehearsal.

RESULTS

Rehearsed on a clone of the grading machine

- 5/5 on the organizers' official validation tasks, at zero Fireworks tokens.
- 19/19 on judge-scale rehearsals across a 2x hardware speed envelope.
- 97.5-100% strict accuracy on held-out tasks, run after run.
- Wall clock 231-380 seconds against a 600-second limit, at every speed.

RESILIENCE, PROVEN THE HARD WAY

Four days of infrastructure storms

- Survived scoring outages, queue stalls, and slow grading hardware.
- Connection retries, a half-open circuit breaker, and voted local fallback
- mean a failed remote call never returns a blank answer.
- Gemma-ready: the probe prefers Gemma when the environment serves it;
- the scoring proxy never exposed a deployment, verified empirically.

Built solo, in five days, in public

- AMD GPU pod + GCP for development. llama.cpp + Qwen3 + Fireworks.
- Reproducible: harness, probe, tuning tools, and rehearsal protocol
- all ship in the repo.
- github.com/DavidOrtsac/frugal-router | MIT license